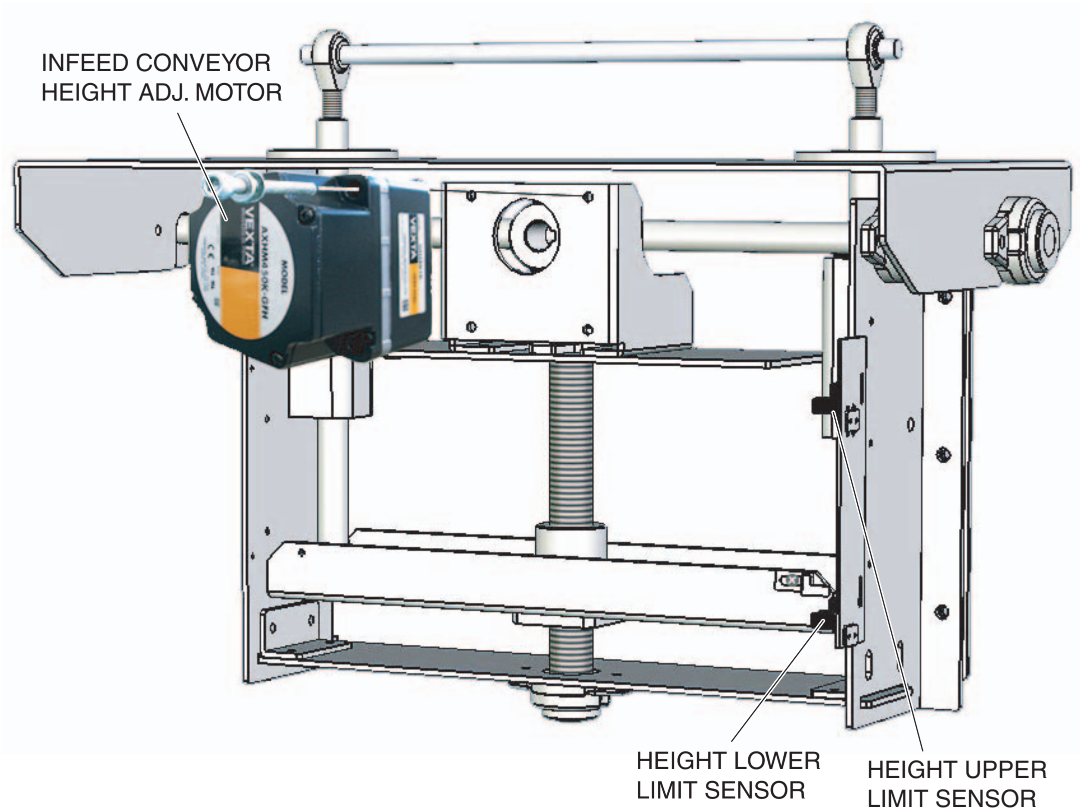
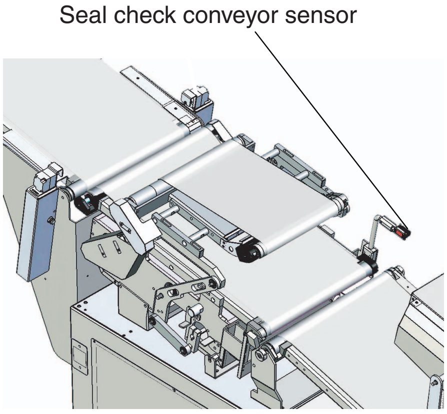
2 REPLACEMENT AND ADJUSTMENT

2.1 Sensor Replacement

2.1.1 Position of each Sensor / Motor in Infeed Conveyor Height ADJ. Unit



2.1.2 Seal check Conveyor Sensor



1. Remove the sensor connector.
2. Remove two hex nuts fixing the sensor with bracket.
3. remove two bolts.
4. replace the sensor with new one.
5. Reinstall all parts in the reverse order of removal.

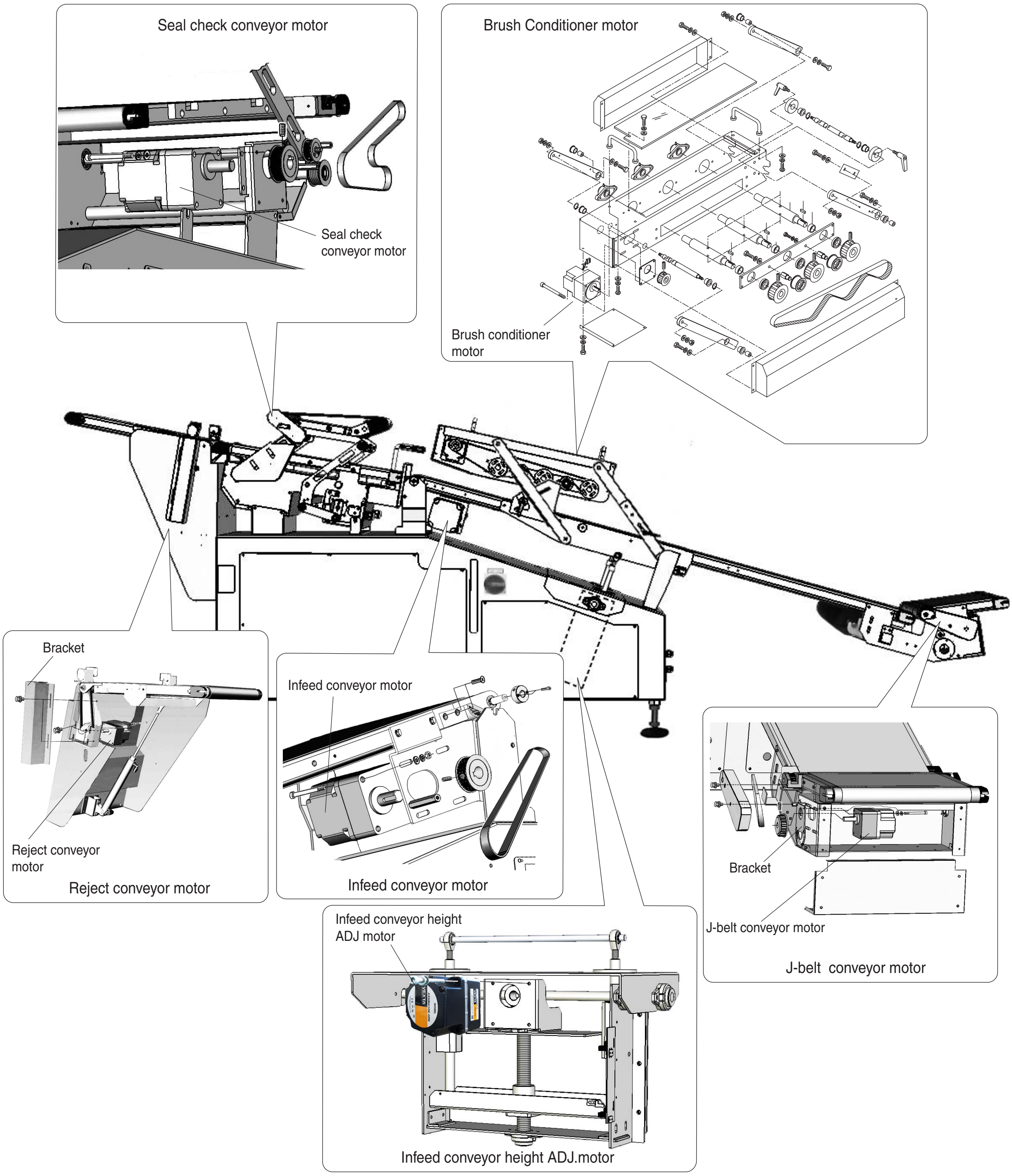
2.2 DC Motors

- J-belt conveyor motor
- Infeed conveyor cotor
- Brush conditioner mortor
- Bag stabilizer front motor
- Bag stabilizer rear motor
- Seal check conveyor motor
- Reject conveyor motor

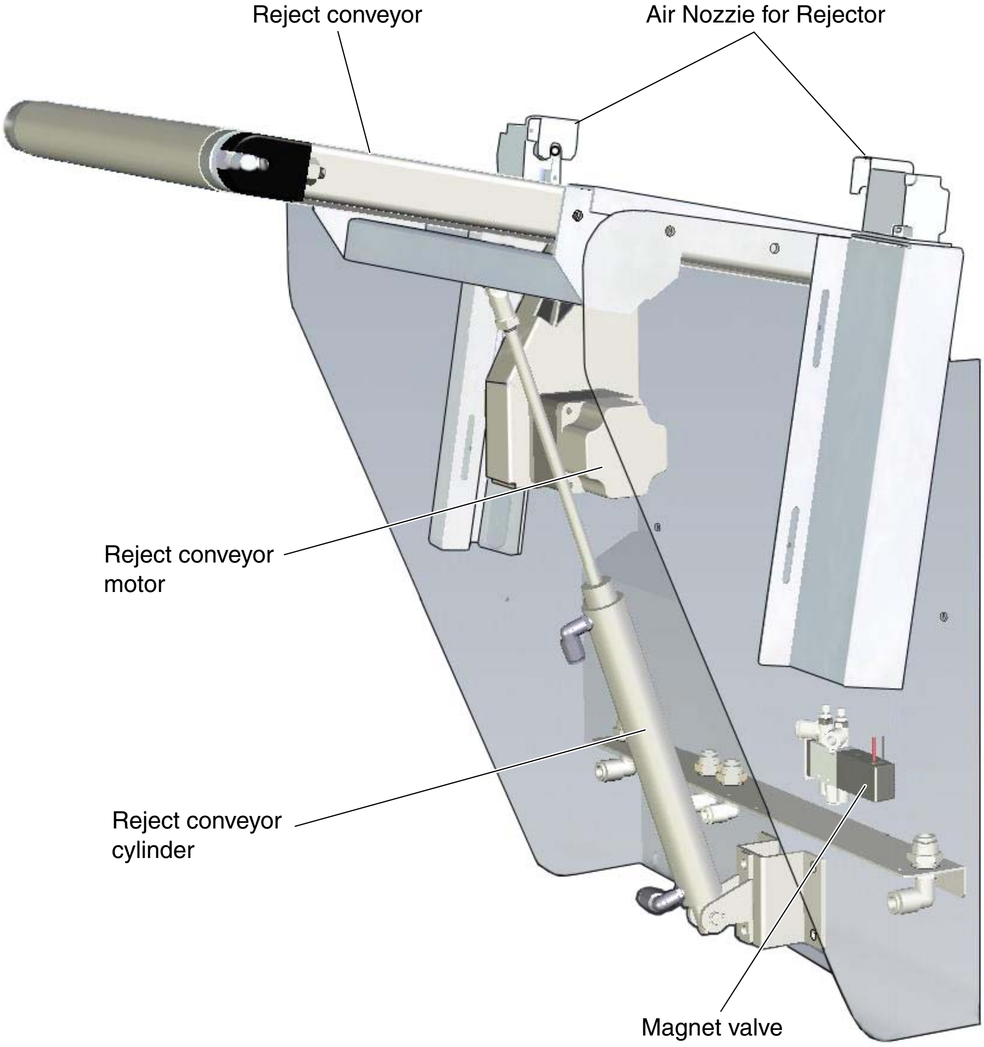
2.2.1 DC Motor Replacement

The replacement of DC motors are basically the same.
Before disassembling, remove the motor connectors and motor pulleys.

1. Remove four screws and remove the motor from motor bracket.
2. Replace the motor.
3. Reinstall all parts in the reverse order of removal.

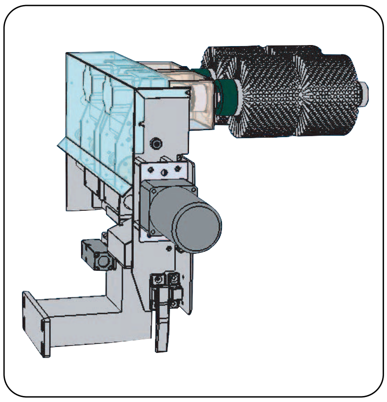

Seal check conveyor motor
Seal check
conveyor motor
Brush Conditioner motor
Brush conditioner
motor
Bracket
Reject conveyor
motor
Reject conveyor motor
Infeed conveyor motor
Infeed conveyor motor
Bracket
J-belt conveyor motor
J-belt conveyor motor
Infeed conveyor height
ADJ motor
Infeed conveyor height ADJ.motor

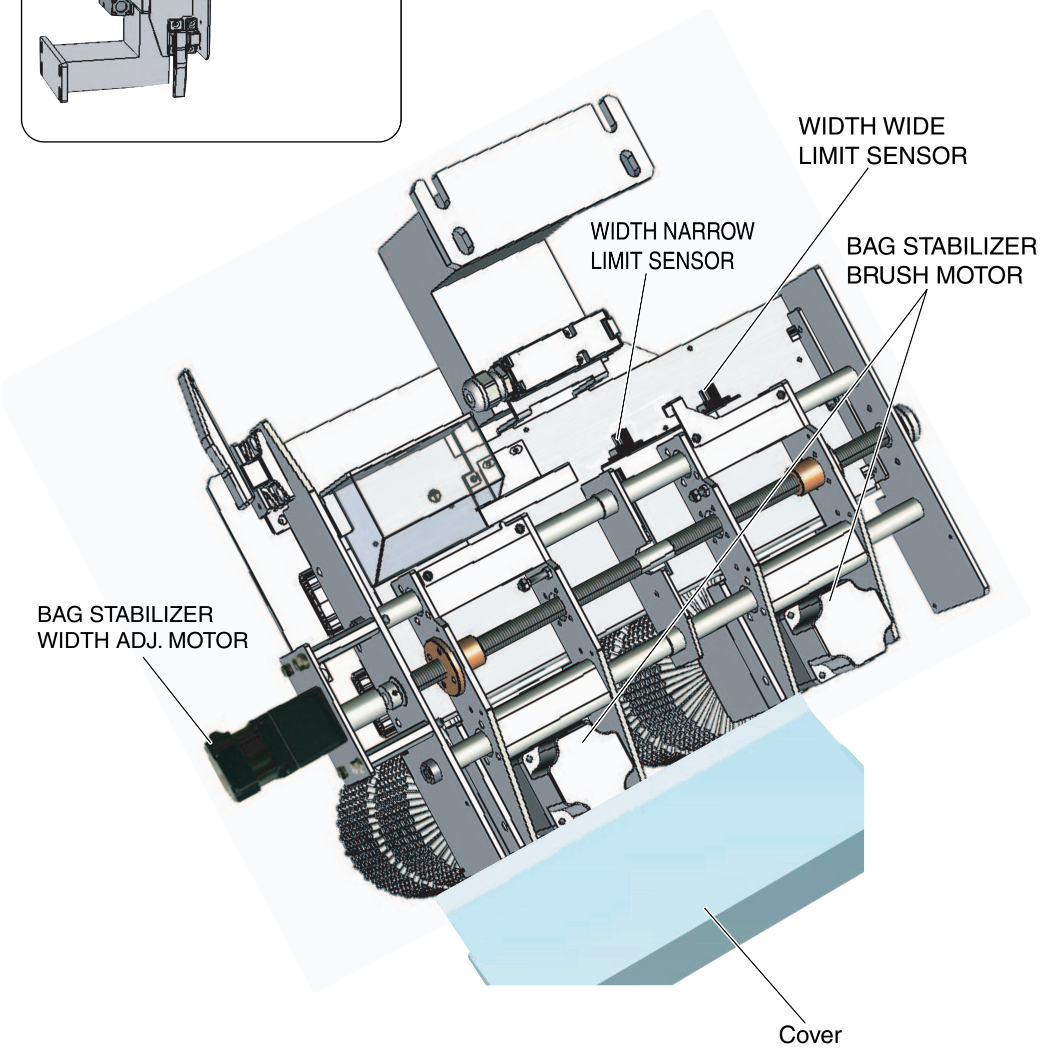
2.2.2 Reject Unit



2.2.3 Bag Stabilizer Unit (Option)

2.2.3.1 Position of each Sensor / Motor in New Stabilizer Unit





2.2.4 Operation (HBL Series DC Motor)

2.2.4.1 Operating the motor

After connection, STOP input or BRAKE input is changed to H (OFF) and after turning the power on, operate the motor according to the mode table for signal input.

Table 2-1 Mode table for signal input

Mode Signal input	CW/CCW	START/STOP	BRAKE
Clockwise (CW) rotation	L	L	L
Counterclockwise (CCW) rotation	H	L	L
Brake (Instantaneous stop)	H or L	L	H
Stop (natural stop)	H or L	H	H or L

*L: ON, H: OFF

Direction of rotation, instantaneous stops and natural stops

- Direction of rotation

  To rotate the motor in a clockwise (CW) direction, CW/CCW input is changed to L (ON). To rotate it in a counterclockwise (CCW) direction, CW/CCW input is changed to H (OFF).

- Instantaneous stops

  The motor stops instantly when the BRAKE input is changed to H (OFF).

- Natural stops

  The motor comes to a natural stop when the STOP input is changed to H (OFF).

2.2.4.2 Operating conditions

NOTE

- Keep the motor case temperature to 90°C or less and the rear panel temperature of the driver to 80°C or less.

Be sure the following conditions apply when running the motor. Motor conditions are defined based on motor temperature rises. The greater the frictional load or inertial load, or the more frequent the start/stops or direction reversals, the higher the temperature will rise.

Conditions for continuous operation

The load torque converted into the motor shaft should not exceed the rated torque.
(See "Speed-torque characteristics," and "Load torque-driver input current characteristics" on page 17 for rated torque, starting torque.)

Speed setting range

The speed converted into the motor shaft should be in the range 300~2,000r/min. Below 300r/min operation is unstable and below about 200r/min the motor stops. Set the speed without load at or above 300r/min.

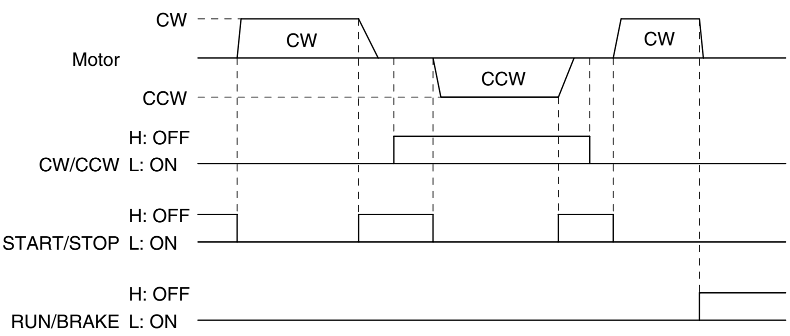
Inertial load conditions

The inertial load (GD2) converted into the motor shaft should be no greater than the permissible load inertia.

Operating patterns

When doing start/stops (instantaneous stops) or direction reversals in short cycles, pay careful attention to motor and driver temperature rises.

2.2.4.3 Example of operational timing chart



The left chart shows an example for change of direction, instantaneous stop and natural stop operations.

2.2.4.4 Protection functions

WARNING

- **When the driver's protection function is triggered, shut off the power immediately. Turn the power back on only after determining the cause. Continuing the operation without determining the cause of the problem may cause malfunction of the motor, leading to injury or damage to equipment.**

The **HBL** series has several protection functions. When they are engaged, an alarm signal is output externally and the motor comes to a natural stop. When the alarm signal is output, shut the DC24V and DC5V power supply off temporarily, eliminate the cause of the alarm, and then turn on the power supply again.

Table 2-2 Protection functions

Type of protection function		Action	Response
Alarm output	Overload protection	This function is engaged when a load in excess of the motorÅfs rated load is applied for roughly five seconds or more or when doing start/stops (instant stops) or direction reversals in short cycles.	Look up the load torque/driver input current characteristics on page 17 and double-check the load torque. If the rated torque has been exceeded, lower the load torque. If the torque is less than or equal to the rated torque, lengthen the operating cycle.
	Open-phase protection	Prevents motor malfunction when the sensor cable within the motor cable is disconnected during motor operation. (An alarm signal will not be output while the motor is at a standstill.)	Check to see if the motor cable connection is broken.

2.2.4.5 Troubleshooting

When the motor is not running properly, try to find the problem in following table. If you still cannot solve the problem, contact your nearest ORIENTAL MOTOR office.

Table 2-3 Troubleshooting

Problem	What to check	Response
Motor does not run.	1. Is the power supply connected correctly?	Connect the power supply correctly.
	2. Are the connector connections secure?	Insert connectors firmly all the way in.
	3. Are the START input and RUN input ON (L level)?	Change the START input and RUN input to ON (L level).
	4. Is the speed potentiometer turned all the way in the counterclockwise direction?	Turn the speed potentiometer to the clockwise. (This applies to both internal and external speed potentiometer.)
	5. Is the alarm signal being output?	Turn the power off, eliminate the cause of the alarm and then turn the power back on after waiting at least five seconds from when the power was turned off. The motor returns to its status prior to the alarm output.
	6. Have you selected the desired speed setting method?	Select the desired speed setting method.
Speed does not change.	1. Have you selected the desired speed setting method?	Select the desired speed setting method.
Motor rotates in an opposite direction.	1. Are you using a gearhead?	Some gearhead speed ratios will cause rotation in the opposite direction to the motor shaft.
	2. Is the CW/CCW input logic correct?	Input the desired rotation direction logic.
Motor does not stop	1. Is it set for STOP input?	Change the STOP input to BRAKE input.
	2. Is the load inertia excessive?	Check the load inertia.

Table 2-3 Troubleshooting (Continued)

Problem	What to check	Response
Alarm is output.	1. Is there an overload?	Check that the load is no greater than the rated load and no greater than the permissible load inertia.
	2. Are direction reversals or start/stops being repeated in short cycles?	Try lengthening the operating cycle or reducing the load.
	3. Is the motor cable connection broken?	If there is anything wrong with the motor cable (sensor signal), the motor will stop. Check for any damage to the motor cable.

2.2.5 Setting the running speed (AXH Series DC Motor)

WARNING

- **Only qualified installers should be assigned to the work of installation, connection, running, operation and inspection.**
  This is intended to prevent fire, electric shock and injury.
- **Before making connections, turn off the driver power.**
  Otherwise, electric shock may occur.

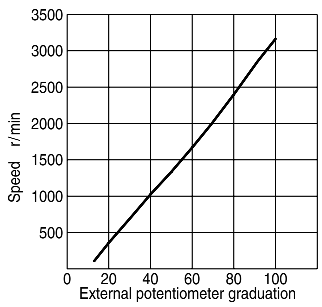
In addition to the driver internal potentiometer, the external potentiometer or external DC voltage can be used to set the motor running speed. The motor speed range is from 100 to 3,000 r/min.
Two running speeds can be set by combining the internal potentiometer and external potentiometer, or the internal potentiometer and external d.c. voltage.

Setting by internal potentiometer

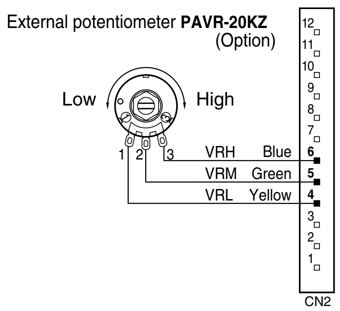
This potentiometer is used when running speed setting is not frequently changed, or when two-step speed switching is performed in combination with external speed setting.
Use a precision screw driver for this adjustment. Clockwise rotation will increase the set speed.
The speed is set to 0 r/min. at time of shipment.
When the motor is driven at the speed set by the internal potentiometer, turn on the INT.VR/ EXT input (low level).
When you want to use only the internal speed setting potentiometer to set the operation speed, you do not have to connect the VRH, VRM and VRL.

Setting by external potentiometer

This potentiometer is used when the speed is set away from the driver or when two-step speed switching is performed in combination with the internal potentiometer.
Use the optional PAVR-20KZ as the external potentiometer.
Clockwise rotation will increase the set speed.



External potentiometer graduation - speed characteristics (typical value)



How to connect the external potentiometer

To drive the motor at the speed set on the external potentiometer, turn off the INT.VR/EXT input (high level).

- To set the running speed only by the external potentiometer, there is no problem if the INT.VR/ EXT input is not connected.

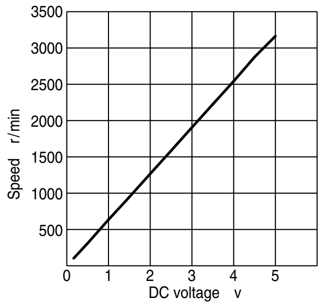
- To perform operation by switching the motor running speed, use the INT.VR/ EXT to switch the external potentiometer and internal potentiometer.

Setting by external d.c. current

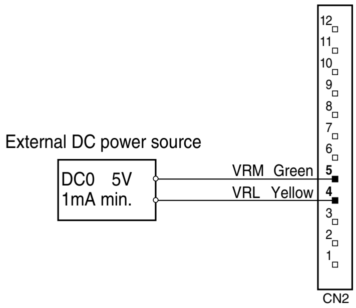
NOTE

- To use a shielded cable for connection with the external potentiometer, connection should be made close to the input/output signal cable connector. Connect the shielded wire to the GND (3).

This method is used to set the speed by the D/A output of a programmable controller or to perform two-step speed switching in combination with the internal potentiometer.
For external d.c. voltage, use d.c. power supply (0 to 5V DC, 1mA or more) where the primary and secondary sides are provided with reinforced insulation.



External DC voltage - speed characteristics



External DC power connection

To drive the motor set at the external d.c.voltage, turn off the INT.VR/EXT input (high level).

- To set the running speed only by the external speed setter, there is no problem if the INT.VR/ EXT input is not connected.
- To perform operation by switching the motor running speed, use the INT.VR/ EXT to switch the external d.c. voltage and internal potentiometer.

NOTE

- The external d.c. power supply voltage must not exceed 5V DC. Otherwise, the driver may be damaged.
- When connecting the external d.c. power supply, sufficient care must be taken not to mistake power po-larity. Connection with incorrect polarity may damage the motor.
- To use a shielded cable for connection with the external d.c. pow-er supply, connection should be made close to the input/output sig-nal cable connector. Connect the shielded wire to the GND (3).

Trouble diagnosis and countermeasures

During motor running, the motor and driver may not operate correctly due to speed setting error or connection error. If normal motor operation cannot be ensured, see the following description and take appropriate countermeasures. If normal operation cannot be ensured even after that, please contact your local sales office.

CAUTION

- **For some time after stopping the equipment, do not touch the motor and the driver by bare hand.**
  You may be burnt by high temperature on the surfaces of the motor and the driver.

Table 2-4 Troubleshooting

Phenomenon	Estimated cause	Measure
The motor fails to turn.	• Either START/ STOP input or RUN/ BRAKE input is not set to the low level.	• Make sure that both START/ STOP input or RUN/ BRAKE input are set to the low level.
	• The internal potentiometer is not adjusted.	• Turn the internal potentiometer slightly in the clockwisedirection. The speed is set to 0r/min. at time of shipment.
	• When the internal potentiometer is used, INT.VR/ EXT input is not set to the low level.	• Set the INT.VR/ EXT input to the low level. When the INT.VR/ EXT input is set to the low level, the internal potentiometer is selected.
	• The external potentiometer contact is faulty.	• Check for connection of the external potentiometer.
	• When the external potentiometer is used, INT.VR/ EXT input is not set to the high level.	• Set the INT.VR/ EXT input to the high level. When the INT.VR/ EXT input is set to the high level, the external potentiometer is selected.
	• The external d.c. voltage contact is faulty	• Check for connection of the external d.c. voltage.
	• When an external d.c. voltage is used, INT.VR/ EXT input is not set to the high level.	• Set the INT.VR/ EXT input to the high level. When the INT.VR/ EXT input is set to the high level, external d.c. voltage is selected.
The motor fails to turn. The motor stops halfway	• Protection function has activated. Count the LED flashings.	• See page 14 and check the causes in conformity to the activated protection function. Take the appropriate measures.
The motor is driven opposite of the specified direction.	• Incorrect CW/CCW input or faulty connection.	• The motors driven in the CW direction when the CW/ CCW input is set to the low level. CCW direction when the CW/ CCW input is set to the high level.
	• Speed reduction ratios 30: 1, 50: 1 and 100: 1 are used in the combination type.	• When speed reduction ratios 30: 1, 50: 1 and 100: 1 are used in the combination type, drive direction is opposite to that of the motor. Reverse the CW/ CCW input operation.

Table 2-4 Troubleshooting (Continued)

Phenomenon	Estimated cause	Measure
Unstable motor operation with big vibration	• The motor (gearhead) output shaft and load shaft are not aligned with each other. • Affected by noise	• Make sure that the motor (gearhead) output shaft and load shaft are connected in an appropriate manner. • Check for running only with the motor, driver and controller required for running. If noise influence has been confirmed, take the appropriate measures such as separation from noise generating source, reconnection of wiring, replacement of the signal cable by a shielded cable, and installation of a Ferrite core.
The motor fails to stop instantaneously.	• The motor is stopped by START/ STOP input.	• Stop the motor by RUN/ BRAKE input.
	• Load inertia may be excessive.	• For this check, increase the frictional load or reduce the load inertia.

2.2.6 Motor rotation direction setting

Setting the dip-sw SW1 on control board P-5530*(☞3.4 P-5530), the direction of the conveyor drive motor can be changed.
Dip-sw SW2 on control board P-5530* must be set according to each specifications like switching Right/Left.

Table 2-5 DIP-SW SW1 Function

SW 1	Conveyor	R type	L type
SW 1-1	Seal check Conveyor	ON	OFF
SW 1-2	Brush conditioner	ON	OFF
SW 1-3	Infeed conveyor	ON	OFF
SW 1-4	Bag stabilizer front	ON	OFF
SW 1-5	J-belt conveyor	OFF	ON
SW1-6	Reject conveyor	ON	OFF

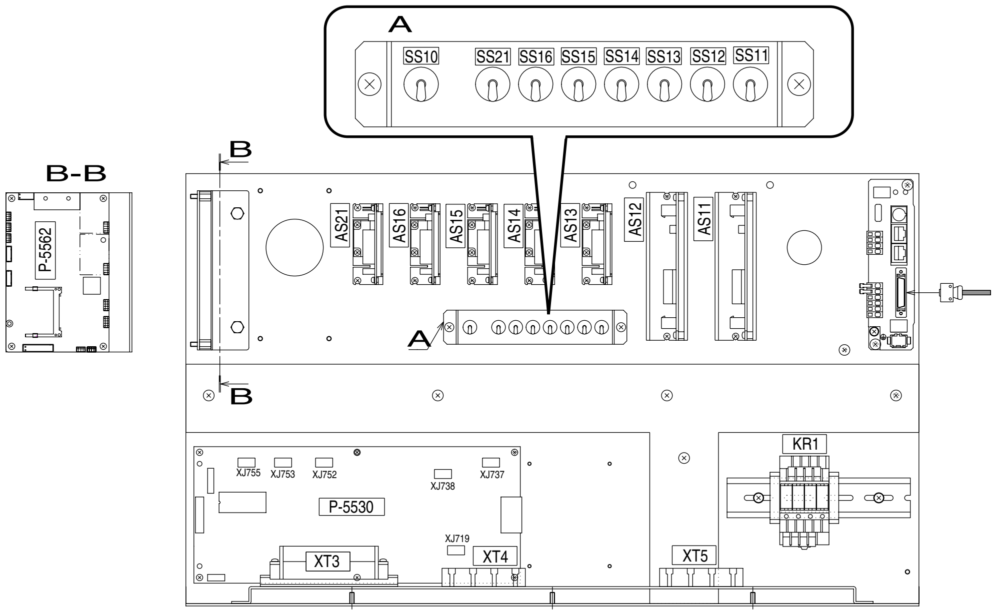
Table 2-6 DIP-SW SW2 Function

SW2	Status	R type	L type	Remarks
SW2-1	Bag stabilizer rear	ON	OFF	-
SW2-2	Downstream Interlock1	—	—	Refer to 5.2
SW2-3	Downstream Interlock2	—	—	Refer to 5.2
SW 2-4	L type	OFF	ON	Set On when L type

2.2.7 Emergency switch

By using the conveyor manual run switch, the conveyor can be manually operated as an emergency measure when the machine does not operate properly due to failure of electric components and so on. In this mode, the machine does not perform seal check but runs only conveyor.

The rotation direction of the motor can be set with the slide switch SW1 on the board. Turning ON SS10, and turn ON the conveyor manual run switch corresponding to each motor. The specified motor starts rotation.



List of motor driver

Part	Manufacturer's model	Conveyor name	Abbreviation	Rotation	Manual run switch	Connector
AS11	HBLD100N	Seal check conveyor	SCC	SW-1	SS11	XJ746
AS12	HBLD100N	Infeed conveyor	IFC	SW-3	SS12	XJ748
AS13	AXHD30K	Brush conditioner	BCD	SW-2	SS13	XJ747
AS14	AXHD30K	Bag stabilizer front	BST	SW-4	SS14	XJ749
AS15	AXHD30K	J-belt conveyor	JBC	SW-5	SS15	XJ750
AS21	AXHD30K	Reject conveyor	RJC	SW-6	SS21	XJ751
AS16	AXHD30K	Bag stabilizer rear	BST	SW-2	SS16	XJ714

*The rotation direction of the motor can be set with the slide switch SW on the board.
*Turn SS10 ON, and turn ON the conveyor manual run switch corresponding to each motor.
The specified motor starts rotation. The speed can be adjusted with the volume on the motor driver.

How to use

1. Open the cover of electric box.
2. Turn ON (upper side) the SS10 switch (A) in the front left.
3. Turn ON (upper side) a set of motor manual run switches from SS11 to SS61 (A) in the front lower right.

With the above procedure, the motor corresponding to each switch of (A) rotates. Adjust the rotation speed with the volume on each motor driver.

As for the reject conveyor, the motor manual run switch and motor drive are located in the reject conveyor motor box. To operate the reject conveyor manual run:

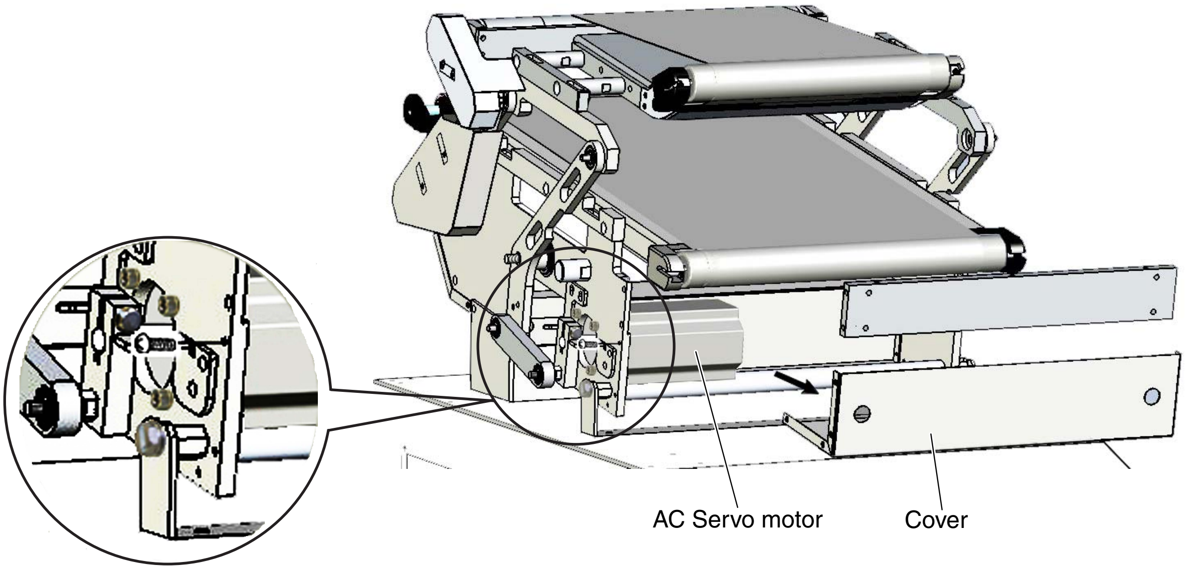
1. Remove the conveyor part from reject conveyor, and remove the top cover of motor box.
2. Turn ON the toggle switch on the motor box.

With the above procedure, the motor of reject conveyor rotates.
Adjust the rotation speed with the volume on the motor driver.

2.3 AC Servo Motor

- AC servo motor for thickness check

2.3.1 AC Servo Motor Replacement



1. Remove one screw from the link of the AC Servo motor.
2. Remove four screws and remove the motor from outer case.
3. Replace the motor.
4. Reinstall all parts in the reverse order of removal.

CAUTION

- **When replacing the AC motor, do not remove the parts (A), (B), (C), and (D) in the figure.**
  When forced to remove the parts, be sure to use the plate when returning them.

2.3.2 AC Servo Parameter Instruction Guide

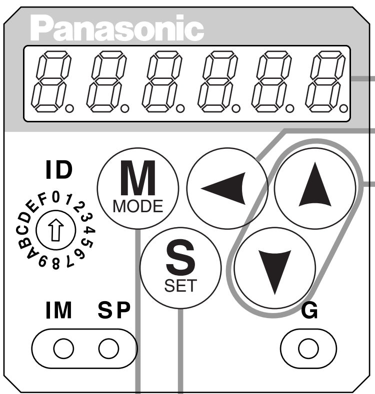
Panasonic®

Technical reference
AC Servo Motor & Driver
MINAS A4-series

- Thank you very much for your purchase of Panasonic AC Servo Motor & Driver, MINAS A4-series.
- Before use, refer this technical reference and safety instructions to ensure proper use. Keep this technical reference and read when necessary.
- Make sure to forward this technical reference for safety to the final user.

If you are the first user of this product, please be sure to purchase and read the optional Engineering Material (DV0P4210), or downloaded Instruction Manual from our Web Site.
[Web address of Motor Company, Matsushita Electric Industrial Co., Ltd.]
http://industrial.panasonic.com/ww/i_e/25000/motor_fa_e/motor_fa_e.html

Setup with the Front Panel



Display LED (6-digit)
All of LED will flash when error occurs,
and switch to error display screen.
All of LED will flash slowly when warning occurs.

Shifting of the digit for data changing to higher digit.
(Valid to the digit whose decimal point flashes.)

Press these to change display and data, select parameters and execute actions.
(Change/Selection/Execution is valid to the digit which decimal point flashes.)
Numerical value increases by pressing ,▲,
decreases by pressing ▼.

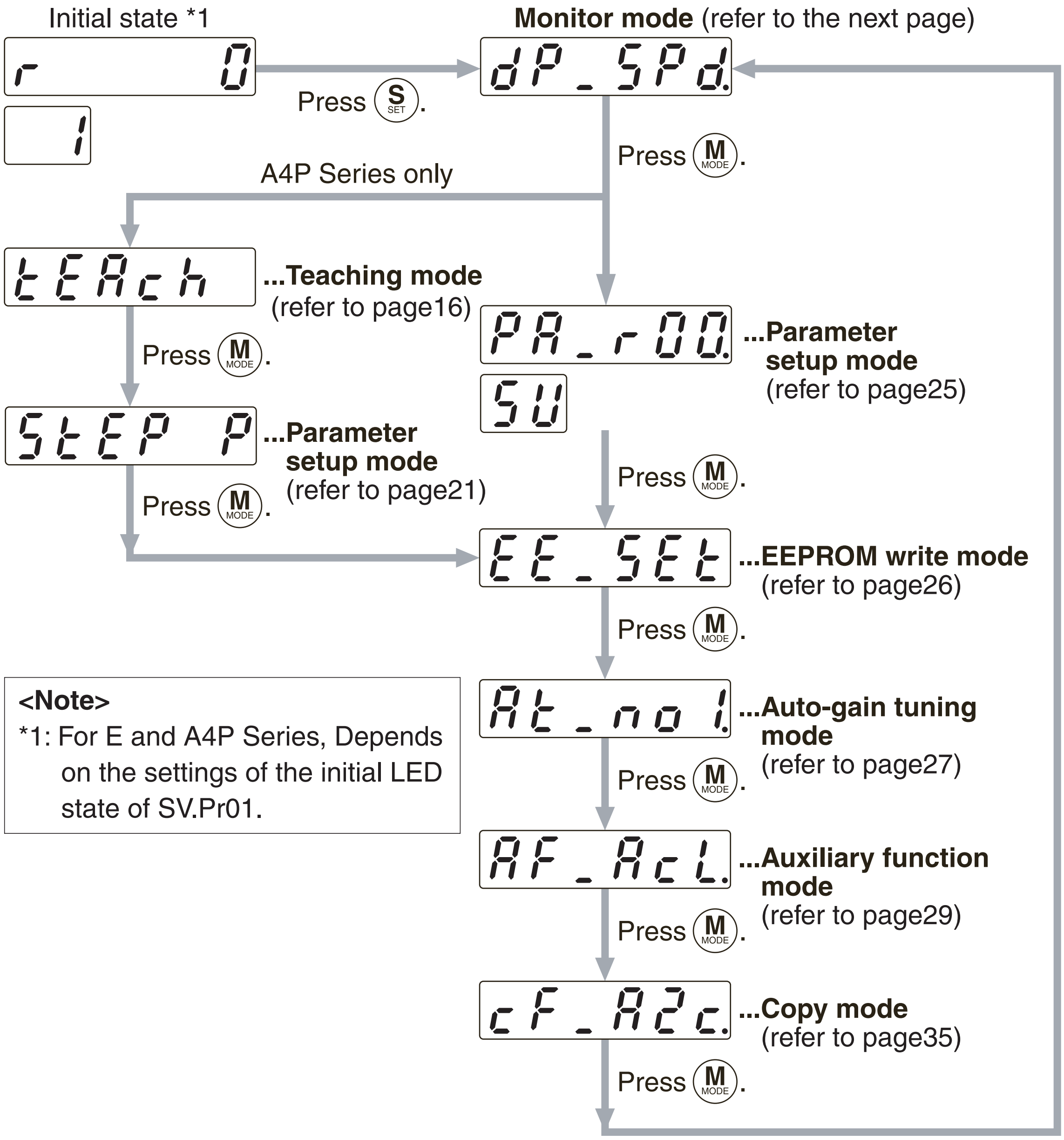
SET Button (valid at any time)
Press this to switch SELECTION and EXECUTTION display.

Mode switching button (valid at SELECTION display)
Press this to switch 5 kinds of mode.
1) Monitor Mode
2) Parameter Set up Mode
3) EEPROM Write Mode
4) Auto-Gain Tuning Mode
5) Auxiliary Function Mode

3. How to Use the Console

Mode Change

The modes below are available in this console. To switch a mode, press (S SET) once in the initial state to enter the **SELECTION display** screen and press (M MODE).



<Note>
*1: For E and A4P Series, Depends on the settings of the initial LED state of SV.Pr01.

Show a target mode to be executed, select it by the (▲)(▼) button and press (S SET) to enter the **EXECUTION display** screen.

4. Monitor Mode

SELECTION display

Display shifts toward the arrowed direction by pressing (▲) and reversed direction by pressing (▼).

EXECUTION display

SELECTION display		⇔ S (SET button)	Display example	Description	Pages to refer
dP_EPS	Positional deviation		P 5	(5 deviation pulses)	P.12
dP_SPd	Motor rotational speed		r 1000	(1000r/min)	P.12
dP_tr9	Torque output		t 100.0	(Torque output 100%)	P.12
dP_cnt	Control mode		PoScnt	(Position control mode)	P.12
dP_ io	I/O signal status		in00. A	(Input signal No.0 : Active)	P.13
dP_Err	Error factor, history		Err. --	(No error currently)	P.14
dP_ no	For manufacturer's use		b.-0 .00		
dP_ rn	Alarm		rn----	(No alarm)	P.14
dP_ rG	Regenerative load factor		rG 30.0	(30% of permissible regenerative power)	P.14
dP_ oL	Overload factor		oL 28.0	(28% of overload factor)	P.14
dP_Jrt	Inertia ratio		J 100	(Inertia ratio 100%)	P.15
dP_nPS	Feedback pulse sum		50	(Feedback pulse sum is 50 pulses.)	P.15
dP_cPS	Command pulse sum		10	(Command pulse sum is 10 pulses.)	P.15
dP_FEr	External scale deviation		FE 0	(External scale deviation is 0 pulses.)	P.15
dP_FPS	External scale feedback pulse sum		0	External scale feedback pulse sum is 0 pulses.	P.15
dP_Aud	Automatic motor recognizing function		Aud on	(Automatic motor recognizing function is validated.)	P.15
dP_ idc	Selection of communication		rS-232	(RS232 communication)	P.15

M MODE (Mode switch button)

Teaching Mode (A4P Series only)	Parameter Mode
P.16	P.21

*1 Not available to the models that do not support external scale.
*2 Not available to MINAS S series.
*3 Only available to MINAS AIII, A4, and A4T.

6. Parameter Setup Mode

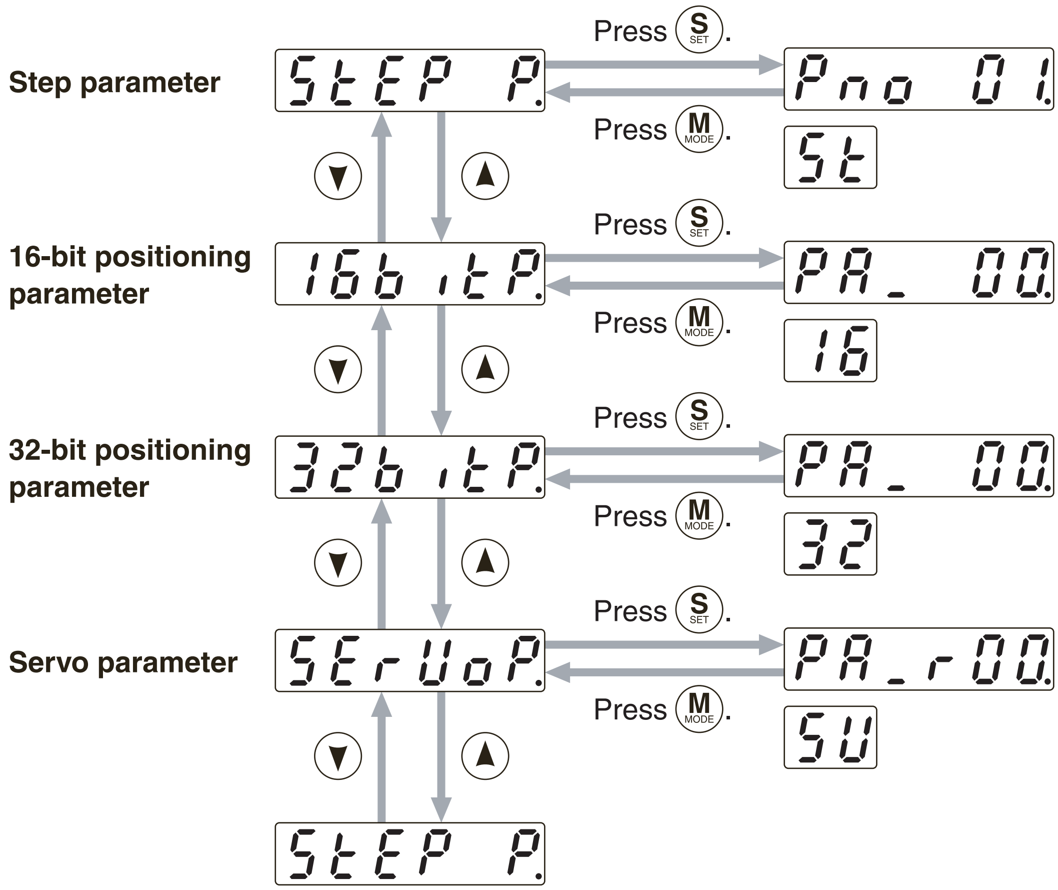
Set the servo driver parameters.
The parameters are classified as follows:

- **Step parameter** (A4P Series only) P.22
- **16-bit positioning parameter** (A4P Series only) P.23
- **32-bit positioning parameter** (A4P Series only) P.24
- **Servo parameter** (All Series) P.25

The parameters that can be set vary depending on the servo driver used.

Structure of Parameter Setup Mode (for A4P Series)

When you press M (MODE) once and S (SET) twice in the initial LED state,
the step parameter display shows StEP P.
Select a target parameter using ▼ and/or ▲.



5. Protective Functions

Protective Function (What Is Error Code ?)

- Various protective functions are equipped in the driver. When these are triggered, the motor will stall due to error, the driver will turn the Servo-Alarm output (ALM) to off (open).
- Error status ands their measures
  - During the error status, the error code No. will be displayed on the front panel LED, and you cannot turn Servo-ON.
  - You can clear the error status by turning on the alarm clear input for 120ms or longer.
  - When overload protection is triggered, you can clear it by turning on the alarm clear signal 10 sec or longer after the error occurs. You can clear the time characteristics by turning off the connection between L1C and L2C or r and t of the control power supply of the driver.
  - You can also clear the above error by operating the "PANATERM®".

<Remarks>

- When the protective function with a prefix of "*" in the protective function table is triggered, you cannot clear with alarm clear input (A-CLR). For resumption, shut off the power to remove the cause of the error and re-enter the power.
- Following errors will not be stored in the error history.

Control power supply under-voltage protection	(Error code No. 11)
Main power supply under-voltage protection	(Error code No. 13)
EEPROM parameter error protection	(Error code No. 36)
EEPROM check code error protection	(Error code No. 37)
Emergency stop input error protection	(Error code No. 39)
External scale auto recognition error protection	(Error code No. 93)
Motor auto recognition error protection	(Error code No. 95)

Warning Function

- The MINAS-A4P Series outputs a warning signal before its protective function is activated, enabling you to check the overload and other error conditions in advance.
  The warning code (about 2 seconds) will alternate at about 4 seconds intervals with 00 on 7-segment LED of front panel when the warning is on.

Warning code No.	Warning function
16	Over-load warning
18	Over-regeneration warning
40	Battery warning
88	Fan-lock warning
89	External scale warning

<Protective function table>

Error code No.	Protective function	Error code No.	Protective function
11	Control power supply under- voltage protection	44	* Absolute single turn counter error protection
12	Over-voltage protection	45	* Absolute multi-turn counter error protection
13	Main power supply under-voltage protection	47	Absolute status error protection
14	* Over-current protection	48	* Encoder Z-phase error protection
15	* Over-heat protection	49	* Encoder CS signal error protection
16	Over-load protection	50	* External scale status 0 error protection
18	* Over-regeneration load protection	51	* External scale status 1 error protection
21	* Encoder communication error protection	52	* External scale status 2 error protection
23	* Encoder communication data error protection	53	* External scale status 3 error protection
24	Position deviation excess protection	54	* External scale status 4 error protection
25	* Hybrid deviation excess error protection	55	* External scale status 5 error protection
26	Over-speed protection	68	Homing error protection
28	* External scale communication data error	69	Undefined data error protection
29	Deviation counter overflow protection	70	* Present position overflow error protection
34	Software limit protection	71	Drive prohibition detection error protection
35	* External scale communication error protection	72	* Maximum movement limit error protection
36	* EEPROM parameter error protection	82	* ID setting error protection
37	* EEPROM check code error protection	93	* External scale auto recognition error protection
39	Emergency stop input error protection	95	* Motor auto recognition error protection
40	Absolute system down error protection	Other No.	* Other error
41	* Absolute counter over error protection		
42	Absolute over-speed error protection		

Seal Checker TSC-RS-120 (Cost-cutting model)
Servo Amp. setting data: using a Panasonic 1/5 gear (200W)

Servo Motor ＭＳＭＤ０２２Ｐ１Ｓ（２００Ｗ）
Servo Amp ＭＡＤＤＴ１２０７（２００Ｗ）
Gear Head ＶＲＳＦ－５Ｂ－２００（１／５）

Servo Amp Parameter

No	Abbreviation	Set up of parameter	Default	Unit	Remarks
02	---	Control mode	0002		Uses in the torque control mode
21	---	Meal time auto tuning set up	0		Ineffective
14	---	1st torque filter time constant	1000	0.01ms	Sets to 10msec
56	---	4th speed setting	6000	r/min	Speed control value
5C	---	Torque command input gain	66	0.1V/100%	Torque reaches 150% in 10 (0.0)V
5E	---	1st torque limit	150	%	
44	---	Numerator of output pulseratio	1500	pulse	Impulses per turn
46	---	Pulse output logic inversion	1		

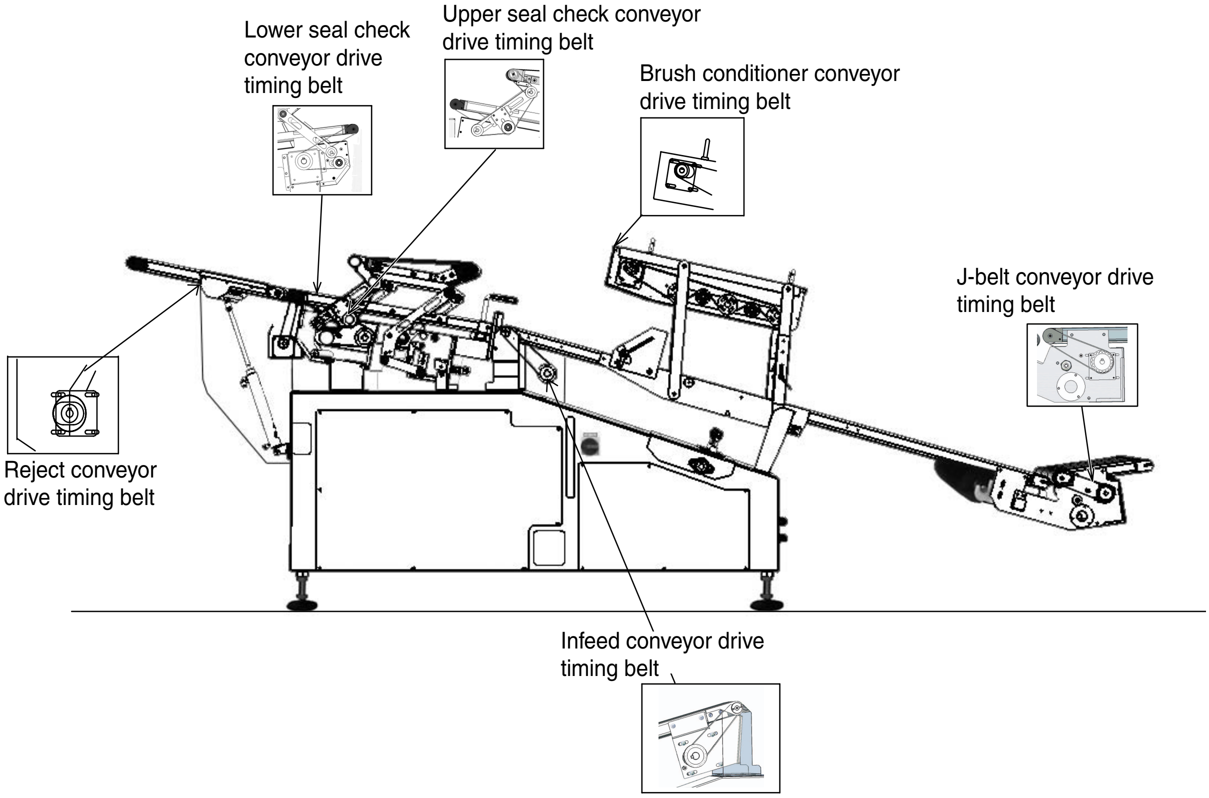
Procedure for setting the servo amp. parameters

① Set the parameters shown in the table above.
② Go to EEPROM save mode to save the EEPROM.
③ Change the main power switch from OFF to ON.

- 1/1 -

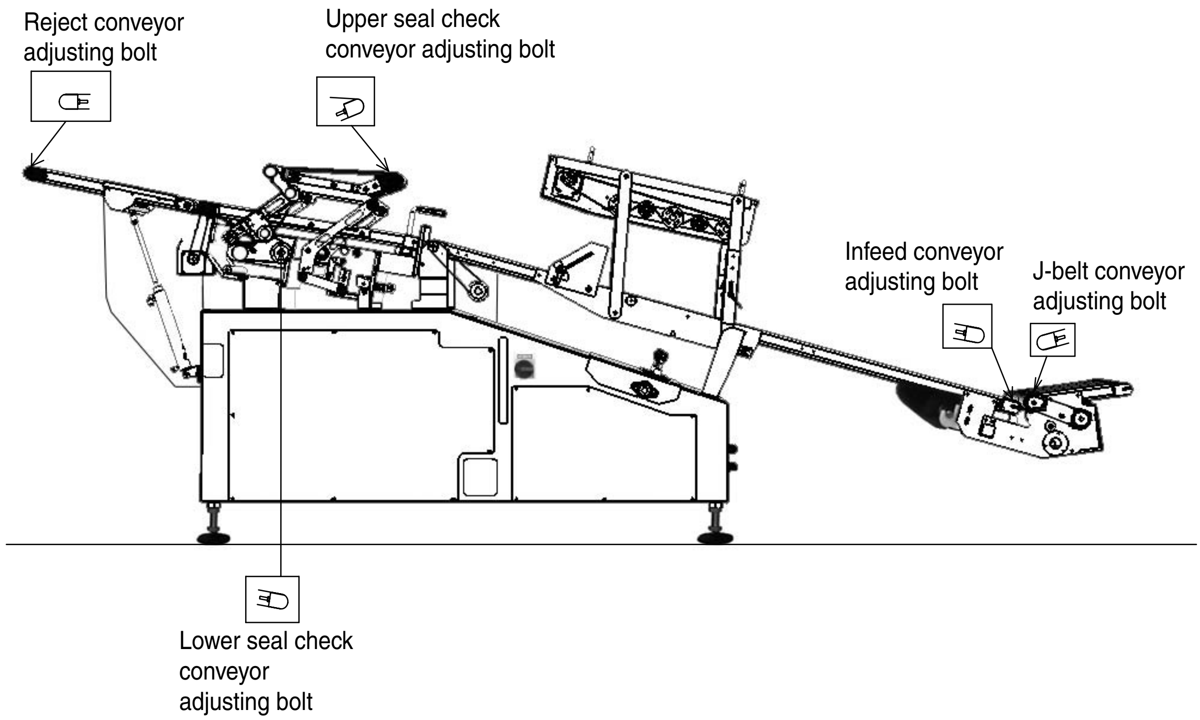
2.4 Timing Belt Tension Adjustment

Adjust the timing belt tension for smooth power transmission and improvement of durable years of driving parts.



- J-belt conveyor drive timing belt
- Infeed conveyor drive timing belt
- Lower seal check conveyor drive timing belt
- Upper seal check conveyor drive timing belt
- Reject conveyor drive timing belt
- Brush conditioner drive timing belt

2.5 Conveyor Belt Tension Adjustment



Adjust the following conveyor belts.

- J-belt conveyor belt
- Infeed conveyor belt
- Lower seal check conveyor belt
- Upper seal check conveyor belt
- Reject conveyor belt

2.6 Test method of seal check conveyor alignment

seal checker TSC-RS-120 measures the height by pinching the object with the upper check conveyor and the lower check conveyor.
Therefore, the check requires that the upper check conveyor and the lower check conveyor should keep parallel.

The following describes how to check the alignment on the check conveyor.

NOTE

- The specified test piece is necessary for this check. Contact our distributor of ISHIDA representative office.

1. Create a new reservation.
2. Set the parameters as follows. Follow the following order for parameter setting:
   a. PROD.SET → PRODUCT

      Product Length = 200mm

   b. PROD.SET → BASIC

      Standby Height = 30mm (Block Height = 35mm) or Standby Height = 37.5mm (Block Height = 42.5mm)

   c. PROD.SET → BASIC → Conveyor Speed Setting

      Seal Check Conveyor Speed 15m/min

      Infeed Conveyor Speed 15m/min

   d. PROD.SET → EXPERT

      Check Start Time = 1550ms
      Check End Time = 1800ms

3. Start the operation to flow the specified test piece for alignment test.
4. Display the waveform graph by pressing the Infotmation key and the ST GRAPH key.
   Press the ZOOM key twice to change to 4X magnification.
5. When the volume of thickness change is within two scales (0.1 mm) in the test time (from Check Start Time to Check End Time), the test succeeds. The green vertical line shows the test time range.
6. In case that the volume of thickness change exceeds the two scales, there is a possibility of slipping of the parallelism between the upper check conveyor and the lower check conveyor. Contact our distributor or ISHIDA representative office.

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<MEMO>